

Task 1: Google Dork

Use Google Search to find:

- 1. PDF files related to Cyber Security**
- 2. PPT files related to Ethical Hacking**
- 3. Cyber Security reports published online**

Write down:

- Search Query Used**
- What Information You Found**

Search Query 1:

filetype:pdf cyber security

Information Found:

- Cyber Security research papers
- Security awareness guides
- Government cyber security frameworks
- Threat intelligence reports

Search Query 2

Query Used:

filetype:ppt ethical hacking

Information Found:

- Ethical hacking presentations
- Penetration testing tutorials
- Network security training slides
- Vulnerability assessment techniques

Search Query 3:

cyber security report filetype:pdf

Information Found:

- Annual cyber threat reports
- Data breach statistics
- Ransomware trend analysis
- Security recommendations for organizations

Task 2: Website Technology Investigation

Visit any 5 websites.

Find:

- 1. Website Name**
- 2. Technologies Used (Frontend, Backend if visible)**
- 3. Is HTTPS enabled?**
- 4. Is Login available?**
- 5. Is Registration available?**
- 6. What security features can you identify?**

Website Name	Frontend Technology	Backend Technology (Visible/Estimated)	HTTPS Enabled	Login Available	Registration Available	Security Features Identified
Google	HTML, CSS, JavaScript	Unknown	Yes	Yes	Yes	HTTPS, Secure Cookies, 2FA
Facebook	React, JavaScript	PHP (estimated)	Yes	Yes	Yes	HTTPS, 2FA, Login Alerts
Instagram	React, JavaScript	Python (estimated)	Yes	Yes	Yes	HTTPS, 2FA
Amazon	JavaScript, HTML, CSS	Java/AWS Services	Yes	Yes	Yes	HTTPS, MFA, Secure Payments
Netflix	React, JavaScript	Java, Node.js	Yes	Yes	Yes	HTTPS, Account Security Controls

Task 3: Email Header Investigation

Open an email in Gmail.

Click:

More → Show Original

Observe:

1. Sender Email
2. Return Path
3. Authentication Results
4. SPF
5. DKIM

Original Message

Message ID	<CACsAOaBFv=DJgXGo328328qwwLUKC=D4Xy1vdXX-JNMT+q9C3w@mail.gmail.com>
Created at:	Fri, Jun 5, 2026 at 10:04 AM (Delivered after 26 seconds)
From:	International Cell GSFCU <international_cell_gsfcu@gsfcuniversity.ac.in>
To:	STUDENTS <students@gsfcuniversity.ac.in>
Subject:	Fwd: EOI_International Summer School Program – HUTECH Global Camp 2026, Vietnam
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain gsfcuniversity-ac-in.20251104.gappssmtp.com Learn more
DMARC:	'PASS' Learn more

Field	Observation
Sender Email	international_cell_gsfcu@gsfcuniversity.ac.in
Return Path	students@gsfcuniversity.ac.in
Authentication Results	PASS
SPF	PASS
DKIM	PASS

SPF (Sender Policy Framework)

- Verifies that the sending server is authorized.

DKIM (DomainKeys Identified Mail)

- Verifies that the email was not modified during transmission.

Authentication Results

- Shows whether SPF and DKIM checks passed or failed.

Task 4: Browser Security Audit

Check your browser and report:

- 1. Number of extensions installed**
- 2. Which extensions have access to all websites?**
- 3. Remove unnecessary extensions**
- 4. Write what security risks unnecessary extensions can create?**

Browser Used:

Google Chrome

Number of Extensions Installed: 5

Extensions with Access to All Websites

Extension Name	Access Level
Grammarly	All Websites
AdBlock	All Websites
Dark Reader	All Websites

Removed Extensions

- Unused Toolbar Extension
- Old PDF Converter Extension

Security Risks of Unnecessary Extensions

1. Can collect browsing history.
2. May steal sensitive information.
3. Can inject advertisements.
4. May track user activity.
5. Could contain malware.
6. Increase browser attack surface.

Task 5: Account Security Check

Review 5 of your online accounts.

Check:

- 1. Is 2FA enabled?**
- 2. Is recovery email configured?**
- 3. Is recovery phone configured?**
- 4. Last password change date**
- 5. Security improvements needed**

Account	2FA Enabled	Recovery Email	Recovery Phone	Last Password Change	Security Improvements Needed
Gmail	Yes	Yes	Yes	Jan 2026	None
Facebook	No	No	Yes	Feb 2026	Stronger password
Instagram	No	Yes	Yes	Mar 2025	Enable 2FA
Amazon	Yes	No	Yes	Dec 2025	Recovery email required
Netflix	No	Yes	No	Aug 2025	Enable 2FA if available

Recommended Security Improvements

- Enable Two-Factor Authentication (2FA) on all accounts.
- Use unique passwords for every account.
- Update passwords regularly.
- Configure recovery email and phone number.
- Review login activity periodically.
- Avoid reusing old passwords.

Conclusion

Through Google Dorking, website analysis, email header inspection, browser auditing, and account security review, various cyber security practices were examined. The investigation highlighted the importance of HTTPS, email authentication (SPF/DKIM), browser extension management, and multi-factor authentication in protecting online accounts and data.